

TidalSense — Data Analyst Technical Challenge

Submission: Please submit a short report and supporting code in the form of a .py file or Jupyter notebook, leaving your working/experimentation untouched (you can comment it out if necessary). You may use any tools you wish to complete this task. **Advice:** You are not expected to find everything – depth of analysis and quality of communication matter just as much as coverage.

Background

TidalSense develops AI-driven diagnostic devices for chronic respiratory conditions. The N-Tidal handset is used by clinicians to perform a breathing test on patients. A diagnostic result is sent and logged on the company's databases.

Over the past 30 days, ten GP practices across a defined region have been participating in a clinical study. Each practice has been issued two handsets and has been conducting tests on patients as part of their normal workflow. Firmware is maintained centrally by TidalSense's software team and updated periodically over the air during the study period; updates take effect the next time a handset is rebooted (normally the start of each day). TidalSense only provides technical support on weekdays during working hours.

You have been provided with two tables:

- **Software table** — one row per test. Contains identifiers for the practice, administrator, and handset, along with error flags, timestamps, boot ID, and the diagnostic result.
- **Hardware telemetry table** — periodic low-level logs transmitted by each handset, covering software version as well as internal hardware components and their status.

Your Task

The clinical and data science teams have two requests:

1. **Study health report** — a brief operational summary of how the study is progressing: test volumes, error rates, diagnostic outcomes, and any notable patterns. Aim for something a non-technical clinical lead could read and act on in a few minutes. [40\%]
2. **Fault investigation** — there are some indications that not everything in the study is running smoothly, whether with the devices, the practices, or the data itself. Investigate, and for each issue you find provide a description of what you found, a likely cause, and a concrete recommendation. [60\%]

This document and the accompanying datasets are confidential and for the purpose of this assessment only. All data are fabricated and do not come from real deployments or real handsets.